

OASYS v3.5.0

Configuration

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1 database credentials

OASYS uses a MariaDB database to store its data. Be aware that MariaDB is no longer fully compatible with MySQL, so a MySQL database cannot be used.

During the installation of OASYS, the database credentials are defined and stored in a configuration file.

If OASYS is installed in the typical way, via the official installer, the database configuration file is located in: `<root folder>/inc/php/db_credentials.php`

In case the database credentials need to be changed, this file can be edited with a text editor.

However for extra security, OASYS also supports the use of environment variables to define the database credentials.

If the following environment variables are defined, they will override the values in the configuration file:

- `db_oasys_host` - the database
- `db_oasys_name` - the database name
- `db_oasys_user` - the database user
- `db_oasys_pass` - the database password

For servers that are running multiple instances of OASYS, it is possible to define custom names for these variables.

You can do so by editing the file `<root folder>/conf/env.ini` and changing the values as you see fit.

2 encryption

For security reasons, OASYS encrypts sensitive data before storing it in the database.

While backend user logins will have hashed passwords, for highest security, other sensitive data such as the password to an LDAP server are encrypted using the OpenSSL library.

Also the passwords for test takers are encrypted before being stored in the database. Since these need to be editable in the editor, respectively exportable, they cannot be hashed.

After installation, no encryption key is defined, and OASYS will generate a random key the first time it tries to read the configuration.

For this reason it is imperative, that the directory `<root folder>/conf/` is writable by the web server user.

Be aware that you should backup the file `<root folder>/conf/crypt.ini` after it has been generated. Without it, parts of your database will be useless and OASYS will not be able to function properly.

Alternatively, you can also define your encryption key in environment variables, by defining the following variables:

- `oasys_aes_key` - the key (32 bytes) written in hexadecimal
- `oasys_aes_iv` - the IV (16 bytes) written in hexadecimal

As with the database credentials, you can also define custom names for these variables by editing the file `<root folder>/conf/env.ini` and changing the values as you see fit.

Superadmins can rotate the database encryption key in System Settings while both frontend and backend maintenance mode are enabled. If `crypt.ini` is active, OASYS generates and activates the replacement key automatically after all encrypted database values have been verified.

For environment-managed keys, the rotation uses temporary pending variables. Their default names are `OASYS_AES_PENDING_KEY` and `OASYS_AES_PENDING_IV`; custom variable names can be configured with `aes_pending_key` and `aes_pending_iv` in `env.ini`. The System Settings dialog guides the administrator through generating the values, reloading PHP, verifying the pending variables, re-encrypting the database, promoting them to the active variables and performing a final verification. Pending key material must be handled as a secret and removed after a completed rotation.